

Education

University of Southern California	Expected May 2027
M.S. in Physics	Los Angeles, CA
University of Minnesota	Sep. 2021 – May 2025
B.S. in Astrophysics and Physics, GPA: 3.54	Minneapolis, MN

Relevant Projects

Gravitational-Wave Astrometry SNR Analysis	Aug. 2025 – Present
– Diagnosed a critical memory bottleneck in a signal-to-noise-ratio simulation (an ~870 GB peak allocation failing at production scale) and redesigned the core algorithm to stream calculations incrementally, cutting peak memory to ~75 MB while validating bit-for-bit numerical equivalence against the original implementation. – Independently formulating and testing hypotheses on signal behavior across varying survey geometries and noise models, running large-scale simulations on USC’s CARC Discovery HPC cluster.	
UltraViolet EXplorer (UVEX)	Mar. 2024 – Jul. 2025
– Built Python simulation pipelines to optimize observation strategies using statistical modeling and resource allocation. – Applied optimization algorithms to maximize event detection under strict time and scheduling constraints. – Collaborated on strategy integration, improving detection efficiency of rare signals and anomalies.	
TURBO Observatory	Jul. 2023 – May 2024
– Engineered low-latency Python pipelines for real-time detection and tracking of objects in under 30 seconds. – Enhanced GitHub-based workflow, debugging Linux pipeline errors and improving reliability.	
S&P 500 Crash Risk Analysis	Mar. 2026 – Present
– Analyzed 90+ years of S&P 500 OHLC data (1927–2026), identifying a data-quality defect affecting 37% of pre-1962 records, including systematic degeneracy during the Great Depression window. – Benchmarked crash-prediction approaches (Linear/Ridge Regression, kNN classification) against a naive random-walk baseline, finding the baseline outperformed regression on test R^2 and that a 95%-accuracy kNN classifier scored $F1 = 0.0$ on crash detection, motivating ongoing tail-risk and bubble-fitting work.	

Work Experience

University of Minnesota Small Satellite Research Laboratory	Nov. 2023 – May 2025
Project Manager	Minneapolis, MN
– Managed development of two NSF/AFRL-funded CubeSats (IMPRESS, EXACT), directing hardware/software integration through Gantt charts and critical-path scheduling under sponsor deadlines. – Held project delivery on schedule through a major design pivot by optimizing resource allocation and task sequencing in real time. – Streamlined testing, task tracking, and documentation practices, improving lab-wide efficiency.	
University of Minnesota	Jan. 2022 – Dec. 2023
Physics Lab Support & Machine Shop Assistant	Minneapolis, MN
– Supported 20+ introductory physics labs, including setup, troubleshooting, and student guidance. – Contributed to technical maintenance, including Helium Liquefier operations and machining tasks.	
Students for the Exploration and Development of Space (SEDS) at UMN	Mar. 2022 – May 2025
Founder & President	Minneapolis, MN
– Founded and scaled a student space organization to 250+ active members, securing \$5,000+ in funding for projects and events.	

Technical Skills

Machine Learning & Statistics: Regression, time-series forecasting, predictive modeling, optimization, signal processing, analytics, Monte Carlo methods	Data Visualization: Matplotlib, Seaborn, Holoviews, Plotly
Programming & Data Analysis: Microsoft Excel, Python (PyTorch, NumPy, SciPy, pandas, scikit-learn, statsmodels), MATLAB, C++, SQL, Linux	Tools & Platforms: Git/GitHub, Jupyter, Linux HPC (Slurm, HTCondor), LaTeX, VS Code
	Workflow & Management: Agile methods, version control, Gantt charts, resource allocation
	AI Tools: Claude Code, Github Copilot, Codex

Publications

A.W. Criswell, S.C. Leggio, M.W. Coughlin, et al. “Electromagnetic follow-up to gravitational wave events with the Ultraviolet Explorer (UVEX).” <i>PASP</i> , vol. 137, no. 5, May 2025. doi:10.1088/1538-3873/adcd9c
L.P. Singer, A.W. Criswell, S.C. Leggio, et al. “Optimal follow-up of gravitational-wave events with the Ultraviolet Explorer (UVEX).” <i>PASP</i> , vol. 137, no. 7, Jul. 2025. doi:10.1088/1538-3873/adcf66